Day 6 – DSA from Zero

Prime Numbers

An easy, beginner-friendly explanation of prime numbers — what they are, how to check them, and how to do it efficiently. Welcome to MK TechVerse 🚀

🚀 DAY 6 | EASY CONCEPT | JAVA CODE

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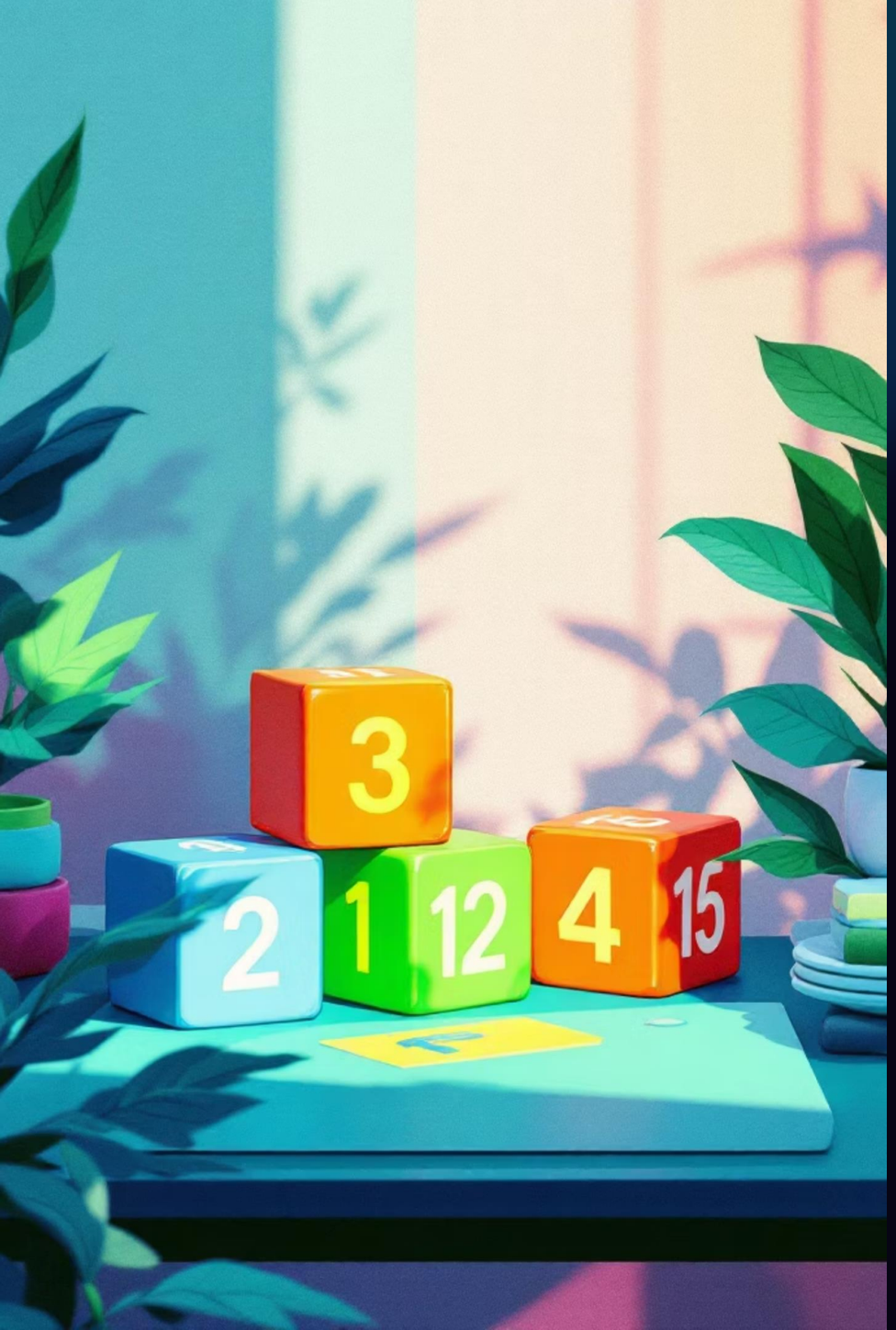

What Is a Prime Number?

A **prime number** is a whole number greater than 1 that can only be divided evenly by **1** and **itself**. No other number divides it without leaving a remainder. Think of it as a number with "no factors" other than the obvious ones.

✅ Prime Numbers

2, 3, 5, 7, 11, 13...

Only divisible by 1 and themselves.

❌ Not Prime (Composite)

4, 6, 8, 9, 10, 12...

Have other divisors besides 1 and themselves.

Special case: The number **1** is *not* prime. The smallest prime number is **2** — and it's the only even prime!

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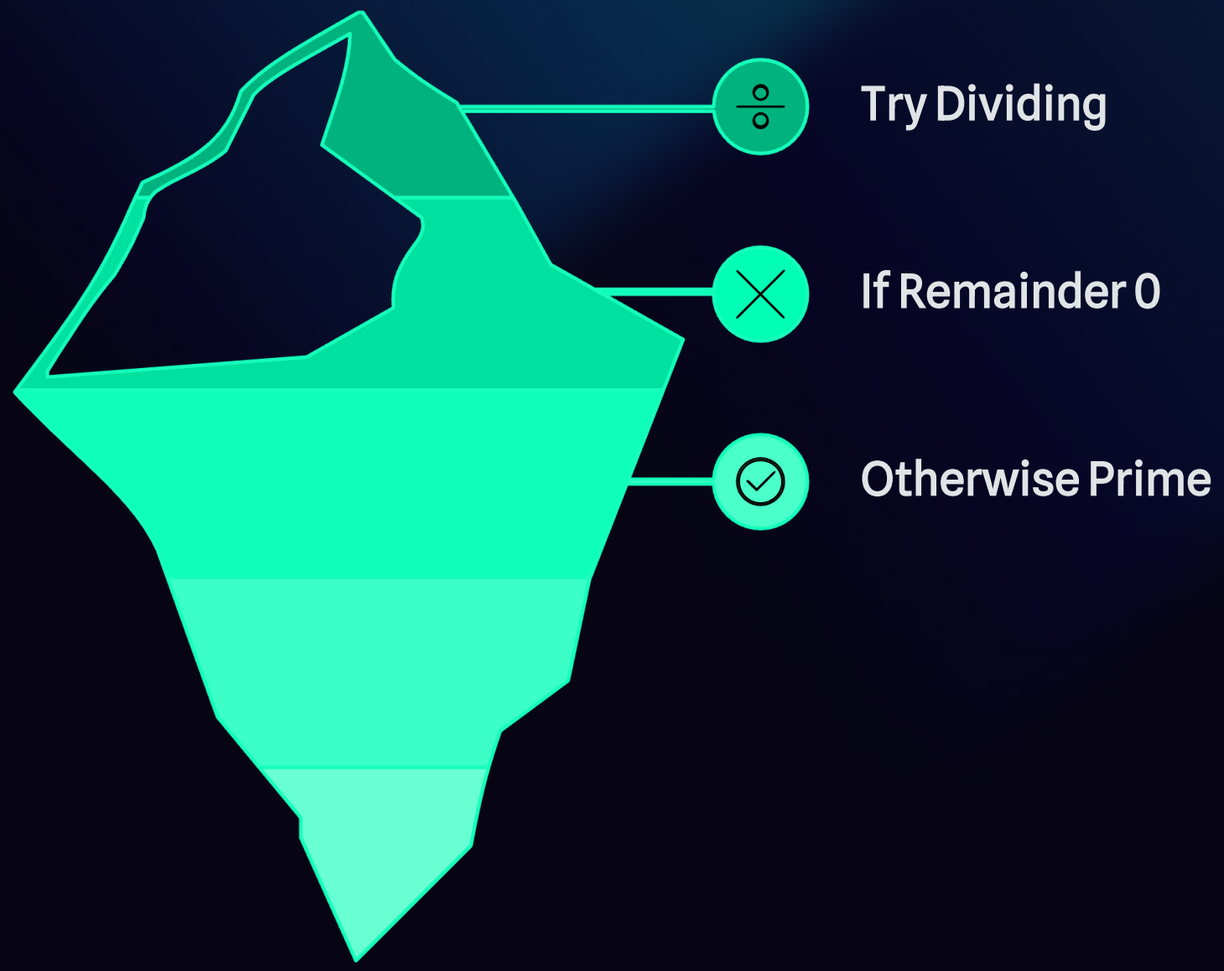
The Simple Idea Behind Checking Primes

Before writing any code, let's build the intuition. The core idea is beautifully simple — try to divide the number by smaller numbers and see what happens.



This trial-division approach is the foundation of every prime-checking algorithm. The key question is: **how far do we need to check?** That's where optimization comes in — and it makes a huge difference in performance.

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Let's Walk Through Examples

Before diving into code, let's manually trace through two numbers to build our intuition. This is exactly the kind of thinking interviewers want to see.

Checking 7 → Prime ✅

Try dividing 7 by numbers starting at 2:

- 7 ÷ 2 = remainder **1** ❌
- 7 ÷ 3 = remainder **1** ❌

Nothing divides evenly → **7 is Prime!**

Checking 8 → Not Prime ❌

Try dividing 8 by numbers starting at 2:

- 8 ÷ 2 = remainder **0** ✅ Divides!

We found a divisor → **8 is Not Prime!** No need to check further.

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Basic Approach — The Slow Way

The most straightforward method is to check **every number** from 2 up to **n – 1**. If any of them divide `n` evenly, it's not prime.

How It Works

Loop from `i = 2` to `i = n-1`. Check if `n % i == 0`. If yes → Not Prime. If the loop finishes with no divisor → Prime.

Time Complexity: O(n)

For `n = 1,000,000`, this does up to **999,999 checks**. That's a lot of work — and it gets worse as `n` grows.

Why It's Not Efficient

We're doing far more work than needed. Many of those checks are redundant. We can do **much better** with a simple mathematical insight.

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Smart Approach — Check Only Up to √n

Here's the key insight: **factors always come in pairs.** If `a × b = n`, then at least one of them must be ≤ √n. So we never need to check beyond √n!

Loop Condition

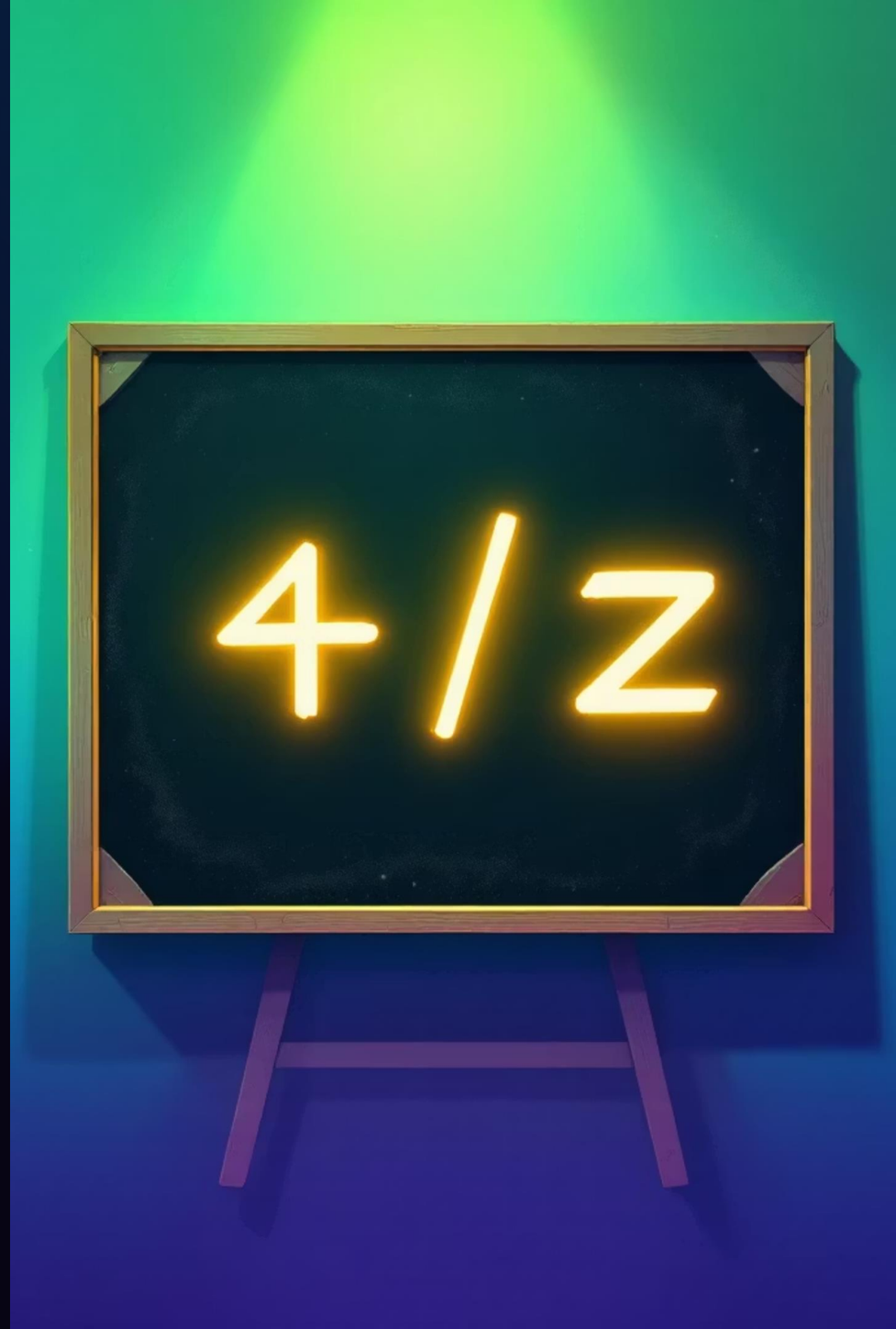
Instead of `i < n`, use `i * i <= n` — this checks up to √n without needing a square root function.

Massive Speedup

For `n = 1,000,000`, we go from **999,999 checks** down to just **1,000 checks**. That's a 1000× improvement!

Time Complexity

O(√n) — dramatically faster than O(n). This is the standard approach used in interviews and competitive programming.



Why √n Works — The Factor Pair Insight

Let's use **36** as our example. Every factor has a "partner" — when you multiply them, you get 36. Notice what happens at $\sqrt{36} = 6$:

Factor Pair	Observation
1 × 36	1 is below $\sqrt{36}$
2 × 18	2 is below $\sqrt{36}$
3 × 12	3 is below $\sqrt{36}$
4 × 9	4 is below $\sqrt{36}$
6 × 6	Both equal $\sqrt{36}$ ← Meeting point!
9 × 4	Repeats — we've seen this!

The pattern: After $\sqrt{n}$, factor pairs just repeat in reverse order. So checking beyond $\sqrt{n}$ is completely unnecessary — we've already found (or ruled out) all possible factors.

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The Code — Java Implementation

Here's the complete, clean Java code using the √n optimization. Notice how `i * i <= n` replaces the need for `Math.sqrt()`.

```
int n = 7;

// Edge case: 1 and below are NOT prime
if (n <= 1) {
    System.out.println("Not Prime");
    return;
}

boolean isPrime = true;

// Check from 2 up to √n (i*i <= n)
for (int i = 2; i * i <= n; i++) {
    if (n % i == 0) {
        isPrime = false;
        break; // Found a divisor, stop early
    }
}

if (isPrime) {
    System.out.println("Prime");
} else {
    System.out.println("Not Prime");
}
```

Key Line

`i * i <= n` — this is the √n optimization. It stops the loop at the square root without calling `Math.sqrt()`.

Why `break`?

Once we find *any* divisor, we know the number is not prime. No need to keep checking — exit early to save time.

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Time & Space Complexity

Understanding complexity is essential for DSA interviews. Here's the full breakdown of our optimized prime-checking solution.

$O(\sqrt{n})$

Time Complexity

We iterate at most $\sqrt{n}$ times. For $n = 10^6$, that's only 1,000 iterations instead of 1,000,000.

$O(1)$

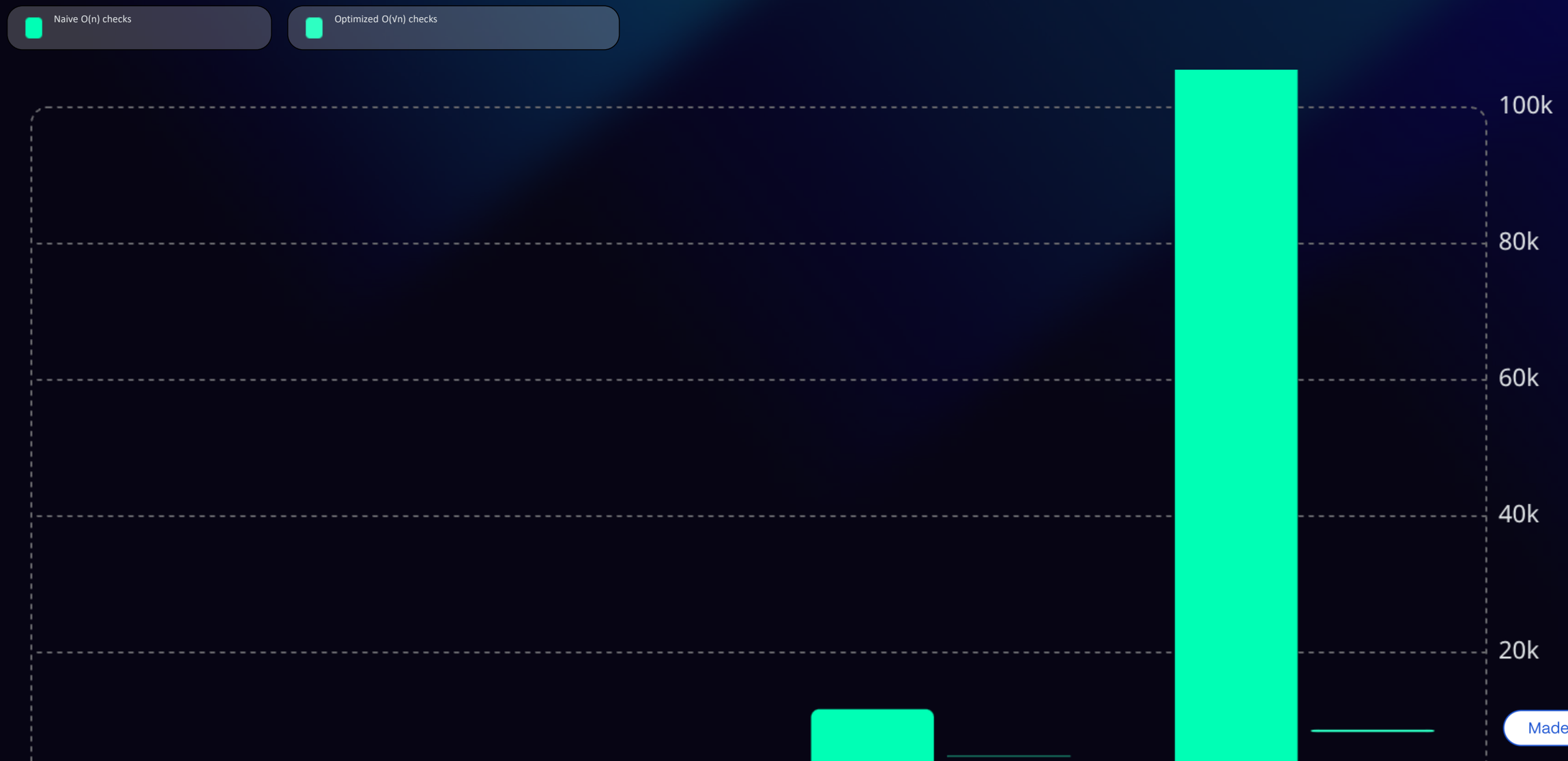
Space Complexity

We only use a single boolean variable `isPrime`. No extra data structures — constant space.

1000×

Speedup vs Naive

Compared to the O(n) approach, the $\sqrt{n}$ optimization gives a massive performance gain for large inputs.



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🏆 Mini Challenge — Try It Yourself!

Before moving on, test your understanding. Trace through the algorithm by hand for these two numbers. Can you determine if they're prime?

1

Check n = 11

$\sqrt{11} \approx 3.3$, so check i = 2 and i = 3.

- 11 ÷ 2 → remainder?
- 11 ÷ 3 → remainder?

Answer: 11 is Prime ✅ — no divisors found!

2

Check n = 12

$\sqrt{12} \approx 3.4$, so check i = 2 and i = 3.

- 12 ÷ 2 → remainder?

Answer: 12 is Not Prime ❌ — divisible by 2!

Homework: Modify the code to print all prime numbers from 1 to 100. This is called the **Sieve of Eratosthenes** concept — we'll cover it soon! 🚀

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